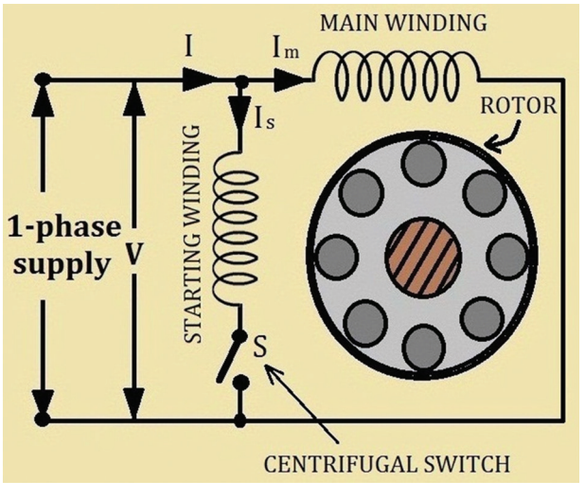


Fig 8.6 *Split Phase Induction Motor*

CONSTRUCTION

The above figure 8.6 shows the schematic diagram of single phase split phase induction motor. The starting winding is designed to have a higher resistance and lower reactance than the main winding. The main winding will have higher inductance when surrounded by more iron which could be made possible by placing it deeper into the stator slots.

WORKING PRINCIPLE

The starting current I_s will lag the main supply voltage V_L by 15° and the main winding current I_m lags main supply voltage by about 80°. Therefore, these currents will differ in time phase and their magnetic fields will combine to produce a rotating magnetic field.

When the motor has come up to about 75% of synchronous speed the starting winding is opened by a centrifugal switch and the motor will continue to operate as a single phase motor. At the point the starting winding is disconnected, the motor develops nearly as much torque with the main winding alone as with both windings connected.

Changing the Direction of Rotation

The direction of rotation of a split phase motor is determined by the way the main winding and auxiliary windings are connected either by changing the main winding terminals or by changing the starting winding terminals, the reversal of direction of rotation could be obtained.

APPLICATIONS

These motors are widely used on easy starting loads of $^1/_3$HP or less

1. Washing machines
2. Driving fans
3. Grinders
4. Wood working tools

b) Capacitor Start Capacitor Run Motor

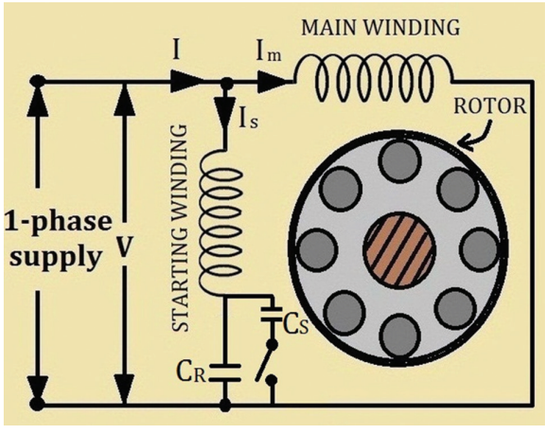


Fig 8.7 *Capacitor Start - Run Motor*

WORKING PRINCIPLE

This motor starts with a high capacitor in series with the starting winding, so that the starting torque is high. When a motor is running, run capacitor is in supply. Both the running and starting windings remain in circuit. At the start, when the centrifugal

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